

2023



Powering Sustainability with Computer Vision on AWS for Opus Insights

Computer-vision startup [Opus Insights](#) migrated to AWS to deliver imaging analytics to clients. Its solution is run from a region that is powered by 100 percent renewable energy and now powers sustainability programs and innovative technologies for agriculture and forestry.

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Scaled

to seven million hectares globally

Unlocks

Provides



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Powers

strategic decision-making



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Computer-vision startup Opus Insights needed compute power at scale, so it migrated its systems to Amazon Web Services (AWS). The company, which was founded in 2018, provides imaging analytics to large forestry and agriculture companies. Its solutions make it simpler to track deforestation, identify risks to crops, and make good on climate pledges.

Opus Insights specializes in processing immense datasets, including high-resolution photos, and turning them into actionable insights tailored to specific business objectives, challenges, and needs. The company works collaboratively with its customers to provide visual evidence of their activities on the ground. This helps companies gain insight into their value chains.



Opportunity | Delivering Advanced Imaging to Power Sustainability Efforts on AWS for Opus Insights

Opus Insights is excited to be building first-of-its-kind imaging technology. “What we’re doing is novel,” says Jaap van der Sluis, CEO of Opus Insights. However, the imaging that it produces calls for intense computing power. To serve multiple clients simultaneously, the company needs a scalable solution that can process many types of data.

Opus Insights started using AWS in 2019. “We were looking for scalability, process parallelization, and managed services,” says Niek Fischer, senior developer at Opus Insights. “Using AWS, we efficiently migrated our data and processing flows from our legacy solution.” After migrating its first service, Opus Insights decided to run all its systems on AWS. It uses [Amazon Elastic Compute Cloud](#) (Amazon EC2), a broad compute service, to process image data at scale with a large number of GPU instances. At the global level, Opus Insights delivers visibility into over seven million hectares of land. Opus Insights runs its solutions from a region that is powered by 100 percent renewable energy.

On AWS, the company supports clients in remote locations all around the world. For example, Opus Insights provided its solution to a mangrove planting project in West Africa. The project aimed to plant mangroves in 10,000 hectares of land. However, using the computer vision of Opus Insights, researchers found that only 26 percent of the promised mangroves had been planted. This critical insight helped them to adjust their go-to-market strategy and their investment controls.

Opus Insights uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service, to retrieve data for clients. By rapidly uploading large quantities of image data to AWS, Opus Insights delivers information to its clients within 48 hours of image capture. The company is also exploring the use of [AWS Outposts](#), which delivers AWS services to virtually any location, to enhance performance in remote places. “Anyone we have set up a camera for can process their imaging rapidly on AWS,” says Van der Sluis. “For us, its speed and scalability are virtually unlimited.”

Outcome | Collaborating Alongside AWS to Deliver Better Services to Global Clients

As a startup that pioneers new technology to assist companies in achieving their environmental stewardship goals, Opus Insights uses AWS support extensively to speed up the problem-solving process. “Working alongside AWS is like going to the grocery store around the corner,” says Van der Sluis. “The convenience of accessing account managers and technical subject matter experts is the greatest benefit of this relationship.”

About Opus Insights

Opus Insights provides innovative computer-vision technology solutions for the net-zero economy and to improve the efficiency and sustainability of agriculture. The Dutch startup was founded in 2018, and it serves customers around the world.

AWS Services Used

Amazon S3



Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

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